Omar Aguilar

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EDUCATION

University of Houston Expected May 2028

Bachelor of Science in Computer Science | Houston, TX

Lone Star College December 2025

Associate of Arts, Computer Science Pathway | GPA: 3.67 | Houston, TX

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript

Web: HTML, CSS, React, Node.js

Tools/Libraries: Git, Linux, Playwright, CMake

PROJECTS

Permit Watch October 2026

Environmental Track Winner | CodeRED Orion Hackathon, University of Houston

React, TypeScript, Node.js, Express, SQLite, MapLibre GL GitHub

- Expanded validated Houston-region permit data from 213 to 807 records by integrating 594 EPA wastewater and TCEQ solid-waste permits with normalization, deduplication, source provenance, and SQLite/Express integration.
- Led React/TypeScript frontend enhancements: URL-preserved 1–10 mile radius search, category/status filters, persistent themes, and MapLibre improvements; fixed map-selection and responsive draft-state bugs with regression tests.
- Collaborated on a four-person team that won the Environmental Track at UH's CodeRED Orion Hackathon with a bilingual environmental-permit discovery platform.

Job Application Automation Platform October 2026 - Present

JavaScript, Node.js, Playwright GitHub

- Built an AI-assisted job application automation platform using Node.js and Playwright with job filtering, saved answers, application history, and a local dashboard.
- Specified and reviewed safeguards that block applications missing required answers, protect prior submission attempts, and preserve interrupted work for explicit retry.
- Used unit/API and synthetic browser tests to investigate failures and verify fixes in supported application workflows.

C++ Console Raycaster January 2025 - February 2025

C++, Windows Console API GitHub

- Implemented a first-person C++ console raycaster using fixed-distance ray stepping over a 16 × 16 map.
- Worked through elapsed-time-based movement, wall-collision checks, distance-based shading, and a map overlay in character-based console output.

WORK EXPERIENCE

Digital Personal Shopper | Walmart January 2026 - August 2026

- Handled 100+ online orders individually during busy shifts using handheld order-management systems.
- Coordinated order fulfillment, substitutions, staging, and customer pickups.

LEADERSHIP

CodePathways Peer Mentor September 2026 - Present

Code[Coogs] x Cougarrettes

- Selected for the Fall 2026 peer mentorship program, with weekly 30+ minute one-on-one meetings focused on a UH student's academic, technical, and professional development.